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Summary

Embedded DSP and AI Engineer with 9+ years of experience in signal processing, low-level optimization, and cross-platform deployment. Expertise in fixed-point implementations, SIMD optimizations (ARM NEON, Qualcomm Hexagon), and performance tuning across ARM, DSP, and embedded systems. Strong background in deploying deep learning models on edge devices including Qualcomm Hexagon DSP. Proven ability to bridge DSP, systems programming, and AI to build efficient, production-ready solutions.

Key Highlights

- Reduced compute cycles by ~60% through float-based optimizations in DSP pipelines.
- Optimized ARMv8-A performance gap from $2.2\times$ to $1.4\times$ vs ARMv7-A using NEON intrinsics.
- Achieved ~90% accuracy with ResNet18 on CIFAR-10 and deployed across CPU, GPU, and DSP.
- Extensive experience in fixed-point conversion, multirate DSP, and real-time audio processing.
- Strong expertise in cross-compilation, profiling (MCPS), and embedded performance tuning.
- Hands-on experience with AI systems including LLMs, RAG, and multi-modal pipelines.

Projects

ResNet18 Deployment on Qualcomm Hexagon DSP (Zentree Labs Jan 2026 - Present)

- Trained ResNet18 CNN on CIFAR-10 using PyTorch Lightning, achieving ~90% top-1 accuracy.
- Built pipeline: PyTorch $\rightarrow$ ONNX $\rightarrow$ QNN compilation (`qnn-onnx-converter`) $\rightarrow$ shared library generation (`qnn-model-lib-generator`).
- Deployed on Snapdragon 710 across CPU (`libQnnCpu.so`), GPU (`libQnnGpu.so`), DSP (`libQnnDsp.so`).
- Cross-compiled ARM64 host application using Android NDK.
- Developed C++ application using `dlopen/dlsym` for dynamic runtime loading.
- Automated deployment and evaluation across all hardware backends.
- Resolved issues with NumPy/QAIRT compatibility and NDK toolchain.

DSP Optimization and Fixed-Point Implementation (Qualcomm (Capgemini) Dec 2016 - May 2023, Jan 2025 - Mar 2025)

- Converted floating-point algorithms to fixed-point, including implementation of `log10` and `pow` using `log2` and `pow2`.
- Used MATLAB Codegen to generate C code; modified MATLAB scripts to resolve codegen issues and replaced dynamic arrays with fixed-size implementations.
- Developed CSIM APIs for MATLAB-based models.
- Implemented optimized FIR filter with decimation-by-2 in assembly by skipping alternate samples; computed stereo outputs simultaneously using shared coefficients.
- Implemented upsampling followed by FIR filtering.
- Automated build and testing using batch scripts, shell scripts, Python, and CMake.
- Applied DSP concepts including FIR/IIR filters and multirate signal processing.
- Optimized matrix multiplication for AGVC (AI/ML-based speech codec) targeting Intel AVX2.

Audio DSP Optimization and Cross-Platform Performance Tuning(Goodix(Capgemini) Aug 2023 - July 2024)

- Integrated RISC-V DSP library FFT/IFFT into VoiceExperience; validated outputs using MATLAB (RMSE in dB) and Adobe Audition.
- Incorporated floating-point FFT/IFFT (SHARC DSP) into fixed-point CarVoice pipeline via floatfixed conversions.
- Optimized performance-critical loops using floating-point processing, achieving ~60% cycle reduction.
- Optimized CarVoice for ARMv8-A, reducing performance gap vs ARMv7-A from $2.2\times$ to $1.4\times$ using ARM NEON intrinsics (25 functions).
- Developed Python scripts for ARMv7-A vs ARMv8-A flat profile comparison.
- Measured MCPS on Android using ADB and profiled with simpleperf.
- Evaluated performance on Raspberry Pi 4; automated MCPS analysis (min-of-5 runs, average, peak).

Certifications

Domain Scholar – Signal Processing and Communications (NPTEL, 60 Weeks)

- Discrete Time Signal Processing, Multirate DSP, Adaptive Signal Processing
- Applied Linear Algebra, Probability Foundations, Information Theory, Signals and Systems

Mathematics for Machine Learning and Data Science – Coursera

- Linear Algebra, Calculus, Probability & Statistics for ML
- A deep understanding of the math that makes machine learning algorithms work.
- Skills acquired : Probability, Bayesian Statistics, Machine Learning Algorithms, Probability Distribution, Machine Learning, Descriptive Statistics, Dimensionality Reduction, Calculus, Sampling (Statistics), Machine Learning Methods, Numerical Analysis, NumPy

Deep Learning Specialization – Coursera

- Designed and trained CNNs, RNNs, and transformer models using TensorFlow
- Applied techniques: hyperparameter tuning, bias-variance analysis, object detection, neural style transfer

ERA V3 – The School of AI, Bangalore

- Gained hands-on experience in Python programming, version control, and web development fundamentals for AI workflows.
- Developed practical expertise in PyTorch, including implementation of neural networks, CNNs, and ResNet architectures, along with optimization techniques.
- Trained transformer models from scratch, covering both fundamental concepts and advanced architectures with efficient training strategies.
- Gained exposure to multi-modal AI systems, AI agents, Model Context Protocol (MCP), and Retrieval-Augmented Generation (RAG) for real-world applications.
- **Capstone: Intelligent Email Query Assistant**
 - Built client-server system with Gmail API OAuth and AWS-hosted LLM backend
 - Enabled natural language email queries with structured search generation
 - Implemented regex filtering, transaction extraction, and MCP-based tool orchestration
 - Developed real-time chat interface using WebSockets

Technical Skills

- **Programming Languages:** C, Python, MATLAB, Assembly (ARM, DSP)
- **Embedded DSP:** Fixed-point implementation, Multirate DSP, FIR/IIR filters, FFT/IFFT, Signal processing algorithms
- **Processor Architectures:** ARMv7-A, ARMv8-A, Qualcomm Hexagon DSP, Intel AVX2, RISC-V DSP
- **Optimization Techniques:** SIMD (ARM NEON), Low-level optimization, Performance tuning, Cycle reduction (MCPS), Profiling (simpleperf)
- **AI Machine Learning:** Deep Learning, CNNs (ResNet), Transformers, LLMs, RAG, Multi-modal systems
- **Frameworks Tools:** PyTorch, PyTorch Lightning, ONNX, QNN SDK
- **Deployment Systems:** Edge AI deployment, Cross-compilation (ARM64, Android NDK), Dynamic linking (dlopen, dlsym)
- **Software Build Tools:** CMake, Git, ADB, Shell scripting, Batch scripting
- **Validation Analysis:** MATLAB, Python scripting, RMSE analysis, Audio validation (Adobe Audition)
- **Platforms:** Windows, Linux, Android, Raspberry Pi